INCH-POUND

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15 December 1992
SUPERSEDING
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11 MAY 1970

MILITARY SPECIFICATION

INDICATORS, SYNCHRO, AIRCRAFT, GENERAL
SPECIFICATION FOR

This specification is approved for use by the Department of the Air Force and is available for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for self-synchronous, remote-indicating type indicators designed to operate on 26 or 115 volt (V), 400-Hz, single-phase power. The nominal stator voltage shall be 11.8V

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Secifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitaion (See 6.2).

SPECIFICATIONS

Federal

PPP-B-636	Boxes, Shipping, Fiberboard
PPP-B-1672	Boxes, Shipping, Reusable With Cushioning

Military

MIL-M-14	Molding Plastics and Molded Parts, Thermosetting

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/TICLA, Tinker AFB, OK 73145-5990 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC: 6620

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MIL-P-116	Preservation, Methods Of
MIL-B-131	Barrier Material, Water-Vaporproof, Flexible Heat Sealable
MIL-T-5350	Transmitter, Synchro Operated, Aircraft, General Specification For
MIL-E-5400	Electronic Equipment, Aerospace, General Specification For
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification For
MIL-A-8625	Anodic Coatings, For Aluminum And Aluminum Alloys
MIL-S-20708	Synchros, General Specification For
MIL-L-25467	Lighting, Integral, Red, Aircraft Instrument, General Specification For
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification For

STANDARDS

Federal

FED-STD-101	Preservation, Packaging And Packing Materials: Test Procedures
FED-STD-595	Colors

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of U. S. Military Property
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-461	Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-810	Environmental Test Methods And Engineering Guidelines
MIL-STD-965	Parts Control Program
MIL-STD-970	Specifications And Standards Order Of Precedence For The Selection Of
MIL-STD-2073/1B	Procedures For Development And Application Of Packaging Requirements
MS3110	Connector, Receptacle, Electric, Series 1, Solder Type, Wall Mounting, Flange, Bayonet Coupling
MS3112	Connector, Receptacle, Electric, Series 1, Box Mounting Bayonet Coupling, Solder Contact

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MS3113	Connector, Receptacle, Electric, Series 1, Solder Type, Solder Mounting, Bayonet Coupling
MS20003	Indicator, Humidity, Card, Three-spot Impregnated Areas
MS27034	Connectors, Receptacle, Electrical, Pin Insert, Cylindrical, Miniature Hermetic, Solder Mount
MS33558	Numerals And Letters, Aircraft Instrument Dial, Standard Form Of
MS33585	Pointers, Dial, Standard Design Of Aircraft Instrument
MS33639	Cases, Instrument, Clamp-Mounted, Aircraft
MS33678	Connector, Receptacle, Electric, Integral Mounting

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, Bldg 4D, 700 Robbins Ave, Philadephia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U. S. Air Force

43A19172 Transformer Assembly - Autosyn Instrument Tester

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Type of testing. The type of testing (qualification or preproduction) to be accomplished shall be as specified in the detail specification.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance (IAW) with MIL-STD-970.

3.3 Preliminary list of electronic parts. Prior to ordering or fabricating parts for production, a preliminary list of electronic parts IAW MIL-E-5400 shall be furnished to the procuring activity. Whenever practicable, the manufacturer's design shall permit the use of standard electronic parts; however, if nonstandard parts are used, they shall be approved by the procuring activity prior to use.

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Data for approval of nonstandard parts shall be prepared and submitted IAW MIL-STD-965.

3.4 Materials

3.4.1 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the indicator, except where magnetic materials are a design requirement.

3.4.2 Nonferrous materials. Nonferrous materials shall be used for all parts of the indicator, except where ferrous materials are a design requirement.

3.4.3 Metals. Metals shall be corrosion-resistant or suitably treated to resist corrosion due to fuels, salt fog, or atmospheric conditions likely to be met in storage or normal service.

3.4.4 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals shall be IAW MIL-STD-454, requirement 16.

3.4.5 Toxic and corrosive fumes. The indicator shall be of such material and construction that fumes, when burned, or gases that combine with the atmosphere to form acid or corrosive alkali will not be liberated.

3.4.6 Protective treatment. When materials are used in the construction of the indicator that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.5 Design and construction. The indicator shall be designed to provide remote indication of aircraft functions when connected to a synchro transmitter conforming to MIL-T-5350. Electrical connections between indicators and transmitters shall be IAW figures 1 and 2.

3.5.1 Indicator synchro. The indicator synchro shall consist of a salient two-pole wound rotor used as the primary and shall be designed for either 26 or 115V, but not both, 400-Hz, single-phase power, with wye-connected stators as the secondaries. The stators shall be so wound that the amplitudes of the induced voltages measured between any two stator leads will vary approximately sinusoidally with the angular positions of the rotor. The indicator may consist of one or more synchros as specified in the detail specification. Pivots, bearings, and gears shall not bind or shake and shall be as nearly frictionless as possible.

3.5.1.1 Secondary voltage. The maximum open-circuit voltage measured between any two secondary leads shall be 11.8V ±3 percent.

3.5.2 Single indicator. The single indicator shall incorporate one self-synchrono motor geared or otherwise connected to a pointer. No supporting bridge over the dial shall be used to support the shaft.

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3.5.3 Dual indicator. The dual indicator shall incorporate two self-synchronous motors geared or otherwise connected to two concentric pointers. No supporting bridge over the dial shall be used to support the pointer shaft.

3.5.4 Receptacles. Receptacles used on the indicators shall be selected IAW table I or II, as specified (see 6.2).

*TABLE I. Indicator receptacles

Single or Dual	Unsealed Indicators	Sealed Indicators	For New Design
Single indicators Dual indicators	MS33678-14S-2P MS33678-16S-1P	MS33678R-14S-2P MS33678R-16S-1P	MS27034Y14T7P MS27034Y12B12

TABLE II. Indicator receptacles (Navy use)

Single or Dual	Unsealed Indicators	Sealed Indicators	For new Design 1/
Single indicators Dual indicators	MS33678-14S-2P MS33678-16S-1P	MS33678R-14S-2P MS33678R-16S-1P	MS3113H10-6P MS3113H10-10P

1/ For nonsealed instruments, a class E or P connector of the same size and pin designation as MS3110, MS3112, or MS3113 may be used.

3.5.5 Synchro receiver. Any synchro receiver used shall have electrical characteristics compatible with those defined in MIL-S-20708.

3.5.6 Electrical receptacles. The electrical receptacles shall be wired as shown on figures 1 and 2, and figures 3 and 4, as applicable. The receptacles shall be centrally located on the indicator and shall have the polarizing key at the top.

3.5.7 Cases. Cases shall be IAW MS33639, except as specified herein and in the detail specification. The bezel ring shall be made of nonferrous, low-density metal and shall be finished in black, color No. 37038 of FED-STD-595. The front edge of the case shall not extend beyond the outer surface of the cover glass by more than 1/32 inch. The case shall be finished in lusterless black, color No. 37038 of FED-STD-595.

3.5.7.1 Nonhermetically sealed. Nonhermetically sealed cases shall be rain-tight. The body of the cases shall be made of either nonferrous, low density metal or of a smooth surface thermosetting plastic conforming to MIL-M-14.

3.5.7.2 Hermetically sealed. The case shall provide a hermetically sealed enclosure for the internal mechanism of the indicator. The design of the case shall be such that the internal mechanism may be removed from the case, replaced and the case resealed without the use of special tools and fixtures unless they are approved by the procuring activity. Hermetic sealing shall be so accomplished that the seal will not be dependent upon materials which will be

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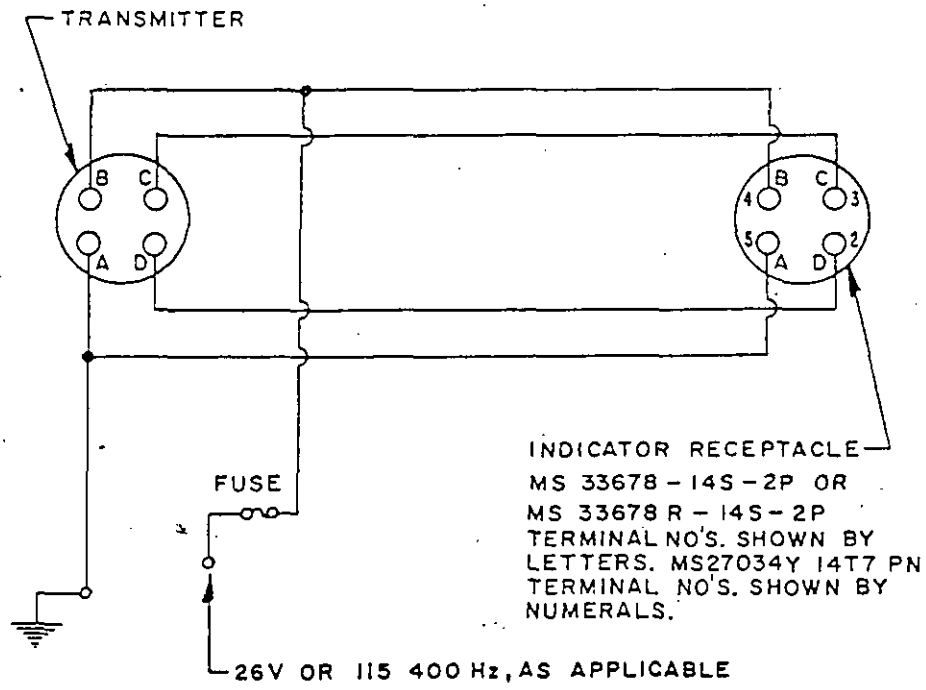


FIGURE 1. Wiring diagram - external, single indicator.

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affected by the action of any atmosphere to which the indicator may be subjected. The case shall be finished in black, color No. 37038 of FED-STD-595.

3.5.7.3 Filling medium. The filling medium shall be of at least 98 percent purity, free of dust particles, and shall contain not more than 0.006 milligram of water vapor per liter (dewpoint -65°C) at the filling pressure. The filling medium shall be a mixture of 88 to 92 percent nitrogen and the remainder helium. Unless otherwise specified in the detail specification, the absolute-pressure of the filling medium in the case shall be 12 to 14.7 pounds per square inch (psi).

3.5.7.4 Filler tube. A tube or other facility for evacuating and filling the case shall be provided at the rear of the case and shall be recessed beneath the level of the back plate surface. If a tube is used, it shall be of such length as to permit three refillings of the case.

3.5.8 Cover glass. The cover glass shall be clear, flat, and free from flaws that would interfere with the normal reading of the instrument. The thickness of the cover glass shall be the minimum practicable and entirely suitable for the purpose.

3.5.8.1 Dial-to-cover glass distance. The distance between the indicator dial and the inside of the cover glass shall be the minimum practicable and shall not exceed 0.125 inch on the single indicator and 0.188 inch on the dual and all lighted indicators.

3.5.9 Dial

3.5.9.1 Fasteners. The dial shall be securely fastened to the frame of the indicator in such a manner that it will not loosen or move with respect to the frame when the indicator is vibrated. If screws are used for attaching the dial they shall be so located that they will not interfere with dial markings or prevent them from being located in the proper position.

3.5.9.2 Markings. All markings shall be so durable as to withstand usage encountered in service. Numerals and letters shall conform to MS33558. Dimensions of markings, numerals, and letters shall be as specified in the detail specification. Unless otherwise specified in the detail specification, the marking finish shall be lusterless white, color No. 37875 of FED-STD-595.

3.5.9.3 Dial finish. The background of the dial, mounting bezel, gaskets, and spacer ring shall be finished in black, color No. 37038 of FED-STD-595

3.5.9.4 Electrical zero reference mark. The dial shall be provided with an electrical zero reference mark on the scale portion equidistant in angular degrees between the first and last dial graduations.

3.5.10 Pointer. Unless otherwise specified, the pointers on nonlighted indicators shall conform to MS33585-8. Pointers for lighted indicators shall be as specified in the detail specification.

3.5.11 Lighting. When required, integral lighting shall be IAW MIL-L-25467 or MIL-L-27160 as specified in the detail specification.

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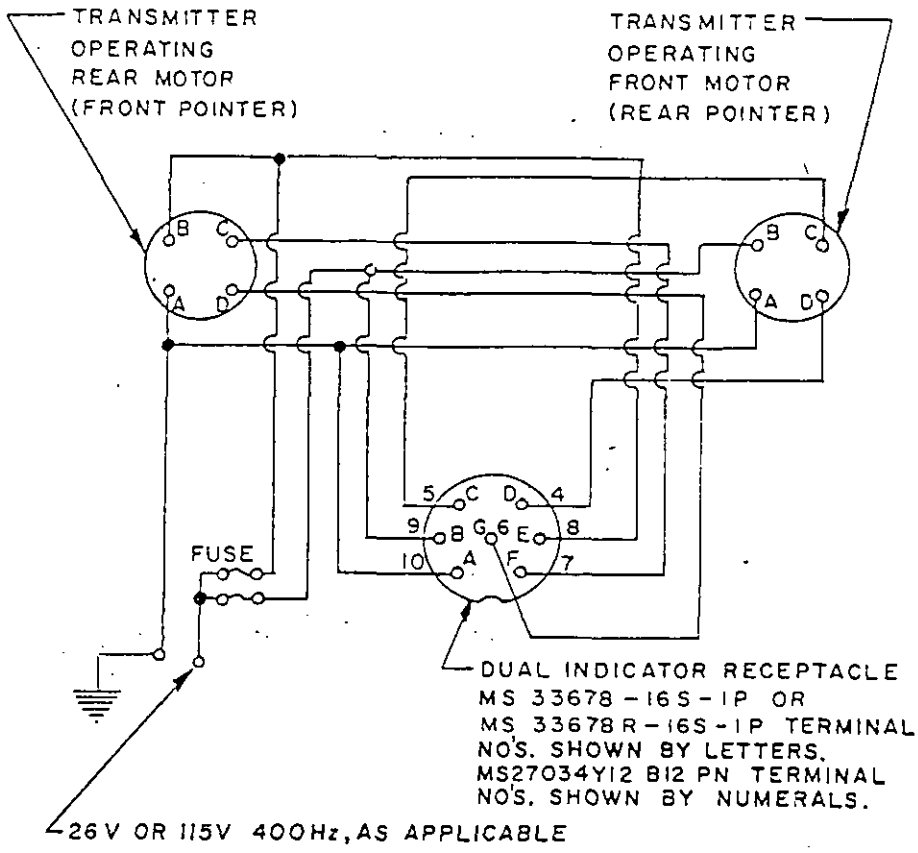
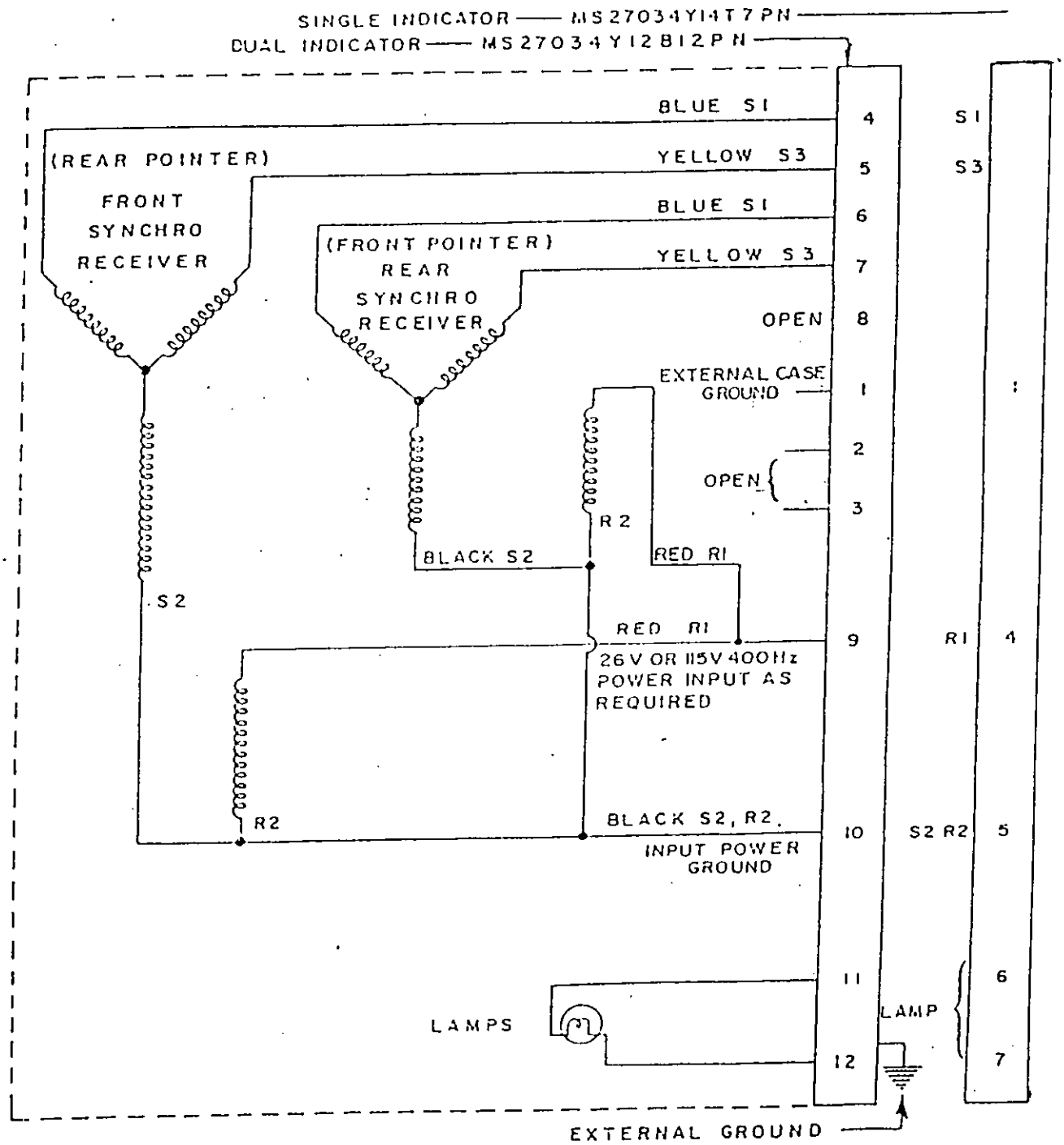


FIGURE 2. Wiring diagram - external, dual indicator.

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FOR UNLIGHTED NEW DESIGN
PINS NUMBERS 11 AND 12 ON DUAL INDICATOR
AND 6 AND 7 ON SINGLE INDICATOR OPEN

*FIGURE 3. Internal wiring - for single and dual lighted indicators.

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3.5.12 Reliability. When reliability is required, it shall be IAW MIL-STD-781 and as specified in the detail specification.

3.6 Performance. The indicator shall operate satisfactorily under the following conditions and those specified in the detail specification.

a. Dielectric strength - withstand a potential of 500V at a commercial frequency for 5 seconds on unsealed indicators and 220V for 5 seconds on hermetically sealed indicators
b. Temperature - temperatures ranging from -65° to $+71^{\circ}$C
c. Vibration - vibration IAW MIL-STD-810, method 514, curve B, except as specified in 4.6.7
d. Altitude - pressure altitudes up to 100,000 feet
e. Humidity - relative humidity up to 95 percent
f. Salt fog - exposure to salt fog for a period of 50 hours
g. Thermal shock - eight cycles of alternate immersion in tap water at a temperature of $85^{\circ} \pm 2^{\circ}$C and then in tap water at $5^{\circ} \pm 2^{\circ}$C (hermetically sealed indicators only)
h. Fungus - fungus growth as encountered in tropical climates (nonhermetically sealed indicators only).

3.6.1 Case leakage (hermetically sealed indicators only). The case leakage shall not exceed 0.1 micron cubic foot per hour.

3.6.2 Effect of lead length. When the lead length between the indicator and transmitter is changed from 5 to 500 feet, the indication shall not change more than 1 degree of arc. The ground lead shall be kept short.

3.6.3 Magnetic effect. Unless otherwise specified in the detail specification, compass deflection shall not exceed 3 degrees of arc when the indicator is operating and positioned 5-1/2 inches from the compass and the horizontal magnetic field intensity at the indicator is 0.17 to 0.19 oersted.

3.6.4 Unit torque gradient. The unit torque gradient shall be a minimum of 2,000 milligram-millimeters per angular degree of rotation.

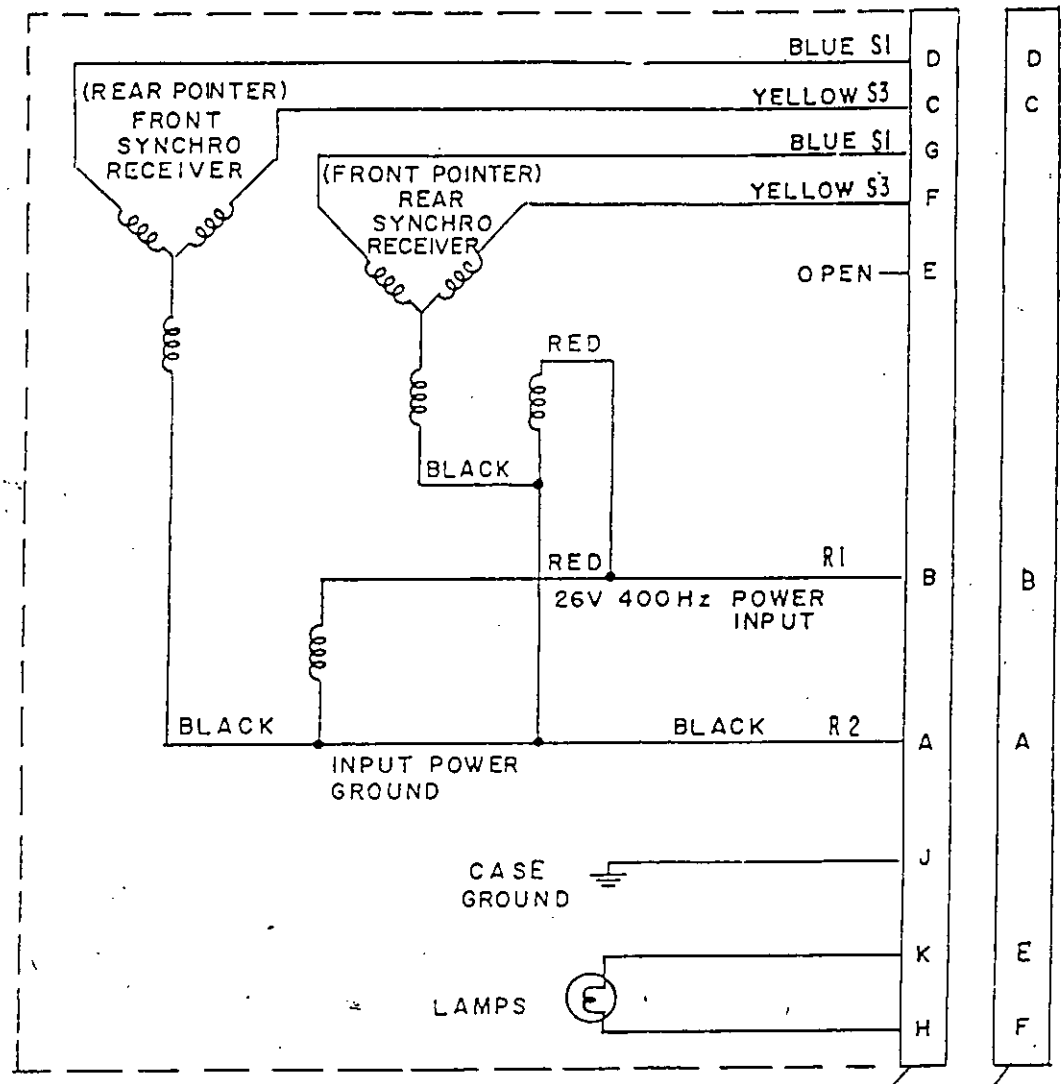
3.6.5 Electromagnetic interference. The indicator shall meet the electromagnetic interference requirements of MIL-STD-461 except the radiated noise shall be from 0.15 to 150 MHz.

3.7 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.8 Internal wiring. Internal wiring shall be IAW figure 3, 4, or 5, as specified (see 6.2). The direction of rotation of the synchro rotor with an increasing input as viewed from the end of the shaft to which the pointer is attached shall be clockwise. S_1 and S_3 leads shown on figures 3 through 6 may be interchanged as necessary to give the increasing indicated function in a clockwise direction.

3.9 Nameplate. A nameplate shall be permanently attached to the rear of the case. Where applicable, the information marked thereon shall conform to

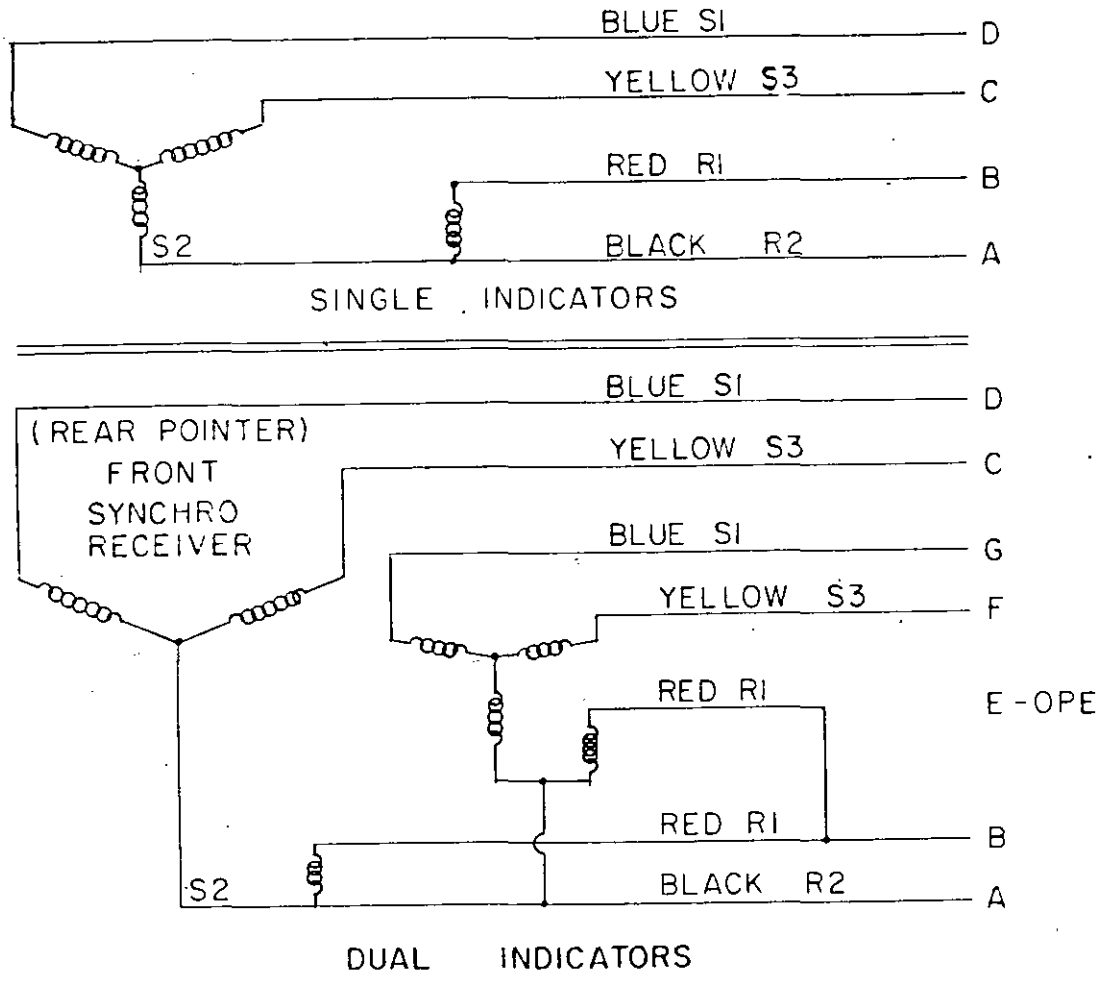
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	DUAL INDICATOR	SINGLE INDICATOR
USE THE CONNECTOR AS REQUIRED LIGHTING AS APPLICABLE. PIN J, IF AVAILABLE, CONNECTS TO CASE.	MS 3113H12-10P OR MS 3112E12-10P MS 3110 12-10P MS 33678-16S-1P MS 33678R-16S-1P	MS3113H10-6P MS 3112 10-6P OR MS3110 10-6P MS 33678-14S-21 MS 33678R-14S-21

FIGURE 4. Internal wiring (navy use).

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NOTE

R1, R2, S1, S2, AND S3 ARE USED AS SPECIFIED IN MIL-S-20708 FOR SYNCHRO TERMINALS.

FIGURE 5. Internal wiring - nonlighted indicators.

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MIL-STD-130, except that Federal stock numbers shall be omitted for Navy use. The nameplate shall not be glued or cemented to the case nor shall metal decals be used.

3.10 Finish

3.10.1 Aluminum alloy parts. Aluminum alloy parts shall be covered with an anodize film conforming to MIL-A-8625, except that the dial, small holes, and case inserts need not be anodized. If such parts are contained within a hermetically sealed enclosure they need not be anodized.

3.11 Dimensions and tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the indicator, they shall be held or limited accordingly.

3.12 Weight. The weight of the indicator shall be as specified in the detail specification.

3.13 Screw threads. Unless otherwise specified, the threads of all machine screw shall conform to MIL-S-7742.

3.14 Workmanship. Workmanship shall be IAW MIL-STD-454, requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification or inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

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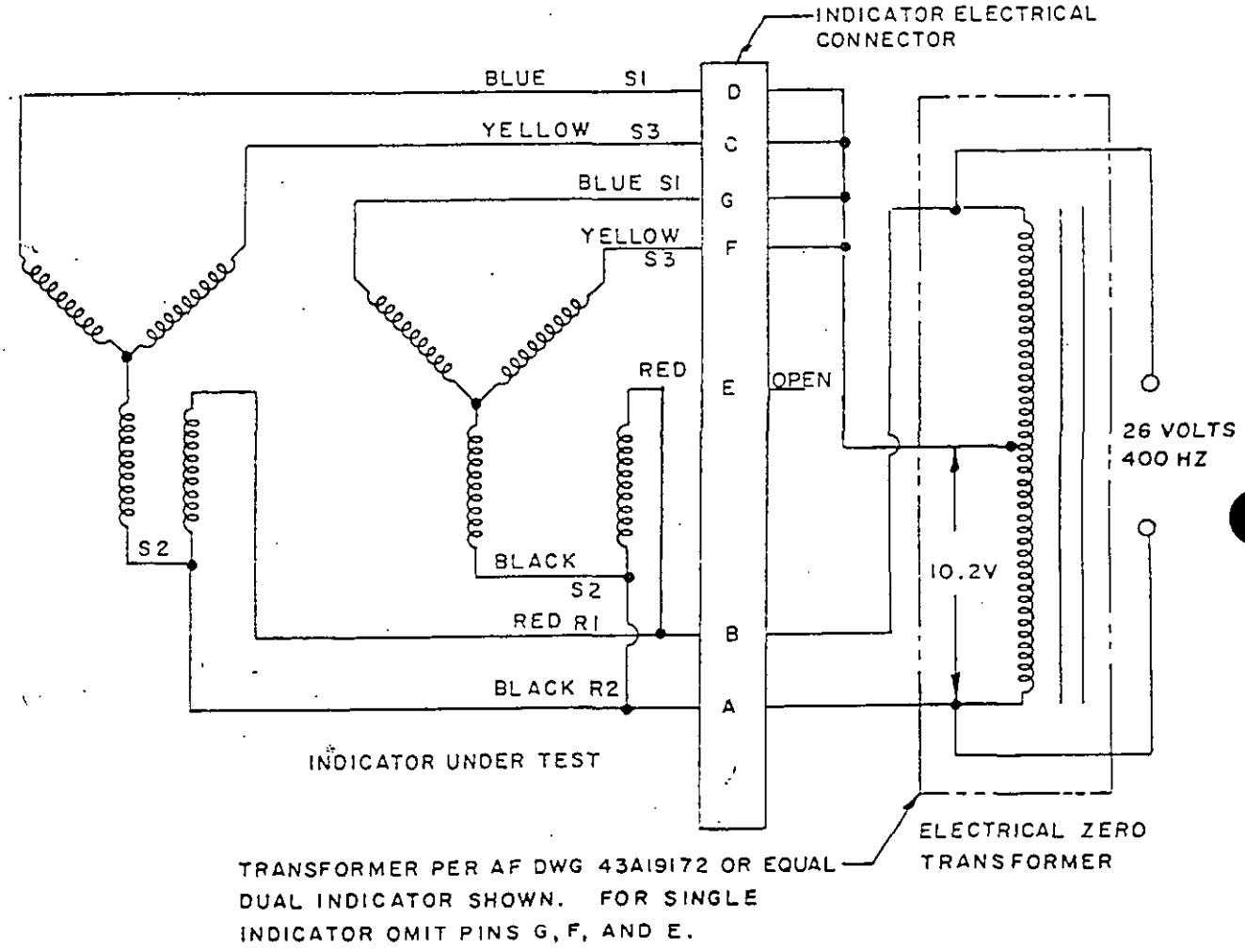


FIGURE 6. Electrical zero test connections.

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4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in (including tests of MIL-L-27160 or MIL-L-25467).

4.3.1 Standard atmospheric conditions. Unless otherwise specified, all tests shall be conducted at atmospheric pressure of from 28 to 32 inches Hg, at a temperature of 24° ±9°C, and at a relative humidity of 90 percent or less.

4.3.2 Attitude. Unless otherwise specified, the indicator shall be tested in the normal operating position with the plane of the dial vertical.

4.3.3 Power supply. Unless otherwise specified, the indicator shall be tested with 26 ±1V 400 ±5 Hz or 115 ±2V 400 ±5 Hz a-c power. On lighted indicators, the lighting input power shall be 4.5 ±0.5V or 27.5 ±1V a-c or d-c power, as applicable.

4.3.4 Electrical zero. Electrical zero shall be determined as specified in 4.6.3.

4.3.5 Test readings. Unless otherwise specified, before a test reading is taken the indicator shall be vibrated at a level of 0.50g ±20 percent.

4.3.6 Test transmitter. A master synchro transmitter having no greater errors than 12 minutes of arc shall be used to test the synchro receiver indicator. The transmitter shall be the same size as the synchro receiver being tested and shall have similar electrical impedances. The transmitter shall be provided with a dial graduated either in 12-or 15-minute increments or in increments of the parameter of the synchro receiver dial which are not more than 20 percent of the smallest receiver indicator tolerance.

4.4 Qualification inspection

4.4.1 Test samples and test report. The test samples and test report shall be as specified in the detail specification.

4.4.2 Qualification inspection. The qualification tests shall consist of all the tests specified under 4.6.

4.5 Quality conformance inspection. Quality conformance tests shall consist of:

a. Individual tests
b. Sampling plans and tests

4.5.1 Individual tests. Each indicator shall be subjected to the following tests as described under 4.6 and in the applicable lighting specification:

a. Examination of product
b. Dielectric strength
c. Individual tests of MIL-L-27160 or MIL-L-25467 (integrally lighted indicators only)
d. Electrical zero
e. Scale error and friction
f. Position error
g. Sealing

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4.5.2 Sampling plans and tests

4.5.2.1 Sampling plan A. One indicator shall be selected at random from each 100 or less produced on the contract or order and subjected to the following tests as described under 4.6 and in the applicable lighting specification:

a. Individual tests
b. Sampling plan A tests of MIL-L-27160 or MIL-L-25467 (integrally lighted indicators only)
c. Vibration error
d. Low temperature
e. High temperature
f. Fogging.

4.5.2.2 Sampling plan B. Unless otherwise specified (see 6.2), three indicators shall be selected at random from the first 15 on the contract or order and subjected to the following tests as specified in 4.6 and in the applicable lighting specification. Tests shall be performed in the order listed.

a. Sampling plan A tests
b. Sampling plan B tests of MIL-L-27160 or MIL-L-25467, except life, contrast, and lamp circuit tests which shall be performed last. (Integrally lighted indicators only)
c. Secondary voltage
d. Magnetic effect
e. Effect of lead length
f. High altitude-low temperature
g. Low temperature exposure
h. High temperature exposure
i. Humidity
j. Thermal shock
k. Fungus
l. Salt fog
m. Electromagnetic interference
n. Unit torque gradient
o. Life, contrast, and lamp circuit.

4.5.2.3 Reject criteria. When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined and corrected. Individual tests may be continued pending the investigation of a sampling test failure. However, final acceptance of items on hand or later produced shall not be made until it is determined that the items meet all the requirements of this specification. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test methods

4.6.1 Examination of product. The indicator shall be inspected to determine compliance with this specification with respect to materials, workmanship, dimensions, weight, scale arc, angles, et cetera. Dials and pointers shall be

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inspected, prior to assembly, as to markings, dimensions, and finishes.

4.6.2 Dielectric strength. If the case is not hermetically sealed, a potential of 500V at a commercial frequency shall be applied between any pin of the electrical receptacle and the case of the indicator for a minimum period of 5 seconds. There shall be no breakdown of the insulation. The same test and tolerance shall apply for hermetically sealed cases except that the potential shall be 200V. This test shall not be conducted on connections that are electrically grounded.

4.6.3 Electrical zero. With the indicator connected and energized as shown on figure 6, the tip of the pointer shall be within 0.010 inch of the electrical zero reference mark on the scale. For dual indicators, the pointer tips shall also be within 0.010 inch of each other. Disconnection at pin D and pin G for dual indicators shall cause the pointer to move approximately 30 electrical degrees in the increasing scale direction. Electrical zero procedures for 115V synchros shall be as specified in the detail specification.

4.6.4 Scale error and friction. The transmitter shall be electrically connected to the indicator under test. The transmitter shaft shall be rotated to provide indications in the test indicator as specified in the detail specification. The test indicator shall be lightly tapped or vibrated before each reading. The difference between the pointer positions before and after the indicator is tapped or vibrated shall constitute the friction error. The difference between the indication or the change made by the transmitter and that of the receiver indicator shall be the scale error. The scale error and friction errors shall not exceed the tolerances specified in the detail specification.. This procedure shall be accomplished by going from the lowest scale graduation of the indicator under test to the highest graduation and return through the scale to the lowest scale graduation. Each unit, in turn, of a multiple-unit indicator, shall be tested for scale error and friction while the remaining unit(s) is set to give a fixed indication. No movement of the unit undergoing scale error and friction tests shall cause a movement of the fixed indication unit(s) in excess of the friction error.

4.6.5 Position error. The indicator pointer shall be positioned at the electrical zero reference mark with the power applied and with electrical connections to the master transmitter. The change in pointer position when the indicator is held in any position on shall not exceed the value specified in the detail specification.

4.6.6 Sealing (hermetically sealed indicators only). The indicator case shall be tested for leaks by means of a mass-spectrometer-type helium leak detector. The detected leak rate shall not exceed 0.1 micron cubic foot per hour at a pressure differential of 12 to 14.7 psi.

4.6.6.1 Sealing (nonhermetically sealed indicators only). The indicator, non-operating, shall be subjected to a rain test in accordance with MIl-STD-810, method 506, procedure I. At the conclusion of this test, the indicator shall meet the test specified in 4.6.4.

4.6.7 Vibration error. The indicator, operating, shall be subjected to a vibration test in accordance with MIL-STD-810, procedure I, curve B, except that time schedule I shall be used. Pointer oscillation and variation shall be as

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specified in the detail specification.

4.6.7.1 Scale error. Following the vibration period, the indicator shall be checked for scale error and friction as specified herein. The scale error and friction shall not exceed room temperature tolerance. No screws or other parts shall become loosened or damaged as a result of vibration.

4.6.8 Low temperature. The indicator shall be subjected to a temperature of -55° ±2°C for a period of 4 hours. While at this temperature, it shall be checked for scale and friction errors as specified in 4.6.4, except that in every case the indicator shall not require more than 3 seconds to come within the tolerance specified. Scale and friction errors shall not exceed the tolerances specified in the detail specification.

4.6.9 High temperature. The indicator shall be subjected to a temperature of 70° ±2°C for a period if 4 hours. While at this temperature, it shall be checked for scale errors and friction as specified in 4.6.4. The scale error and friction tolerances shall not exceed the values specified in the detail specification.

4.6.10 Fogging. The hermetically sealed indicator shall be energized and placed in a 71° ±2°C controlled ambient temperature for a minimum period of 1 hour. After this period and while still at the specified temperature, an ice cube shall be rubbed on the cover glass for a period of 1 to 2 minutes. The glass shall be wiped dry (do not use compressed air) and the indicator inspected for evidence of water or oil fog inside the cover glass. Evidence of fogging shall be cause for rejection.

4.6.11 Secondary voltage. The maximum open-circuit voltage shall be measured between any two secondary leads and shall be 11.8V ±3 percent. The variation between the three maximum secondary voltages on any one indicator synchro as measured between any two stator leads shall not exceed 0.2V.

4.6.12 Magnetic effect. The indicator, first not operating then operating on rated voltage, shall be rotated in a vertical plane about a short-bar magnetic compass. This vertical plane of rotation shall extend through the center of the bar of the magnetic compass and through positions magnetically East and West of the short-bar magnet of the compass. During rotation, the nearest part of the indicator shall be 5-1/2 inches from the center of the compass. Starting directly under the compass at 0°, the indicator shall be rotated clockwise to and held in positions 45°, 90°, 135°, 180°, 225°, 270° and 315° from the original position. At each of these positions, the indicator shall be rotated on its own horizontal axis until it is again in its normal upright position. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. Deflection of the compass at any of the specified positions shall not exceed 3°.

4.6.13 Effect of lead length. The indicator shall not change more than 1 degree of arc when the lead length between the indicator and transmitter is changed from 5 to 500 feet of the No. 20 AWG gage wire. Equivalent resistances may be substituted for the added wire length. The ground lead shall be kept short.

4.6.14 High altitude - low temperature. The high-altitude low-temperature test shall be conducted at a temperature of -55°C and with a chamber pressure

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equivalent to an altitude 100,000 feet for a period of 4 hours. While under these conditions and at the end of the 4-hour period, a scale and friction error test shall be conducted. The scale errors shall not exceed the tolerances specified in the detail specification. The indicator shall not be damaged as a result of this test.

4.6.15 Low temperature exposure. The indicator shall be subjected to a temperature of -65° $+2^{\circ}$C for a period of 48 hours. The temperature shall then be raised to -55° $\pm 2^{\circ}$C and held for a period of 24 hours. While at this temperature, the indicator shall be subjected to the scale error and friction test except that in every case, the indicator shall not require more than 4 seconds to come within the tolerances specified. The scale error and friction shall not exceed the tolerances specified in the detail specification. All parts of the indicator shall then be maintained at room temperature for a period of not less than 4 hours and the indicator again subjected to the scale error and friction test. The scale and friction errors shall not exceed the tolerances specified in the detail specification. There shall be no damage or deterioration that will affect subsequent operation.

4.6.16 High temperature exposure. The indicator shall be subjected to a temperature of 70° $\pm 2^{\circ}$C for a period of 24 hours and then tested at this temperature for scale errors as specified in 4.6.4. The scale and friction errors shall not exceed the tolerances specified in the detail specification. All parts of the indicator shall then be maintained at room temperature for a period of not less than 4 hours and the indicator subjected to the scale error and friction tests. The scale and friction errors observed shall not exceed the tolerances specified in the detail specification. There shall be no damage or deterioration that might affect subsequent operation.

4.6.17 Humidity. This test is applicable only to nonhermetically sealed indicators. The indicator shall be subjected to the scale error and friction test at room temperature then to a humidity test in accordance with MIl-STD-810, method 507, procedure I. External connections shall be made to the indicator to simulate installed conditions. Precautions shall be taken to prevent moisture from dripping on the indicator. Immediately following the completion of the tenth cycle, the indicator shall be subjected to the dielectric strength and scale error and friction tests. The scale readings shall not differ from the readings obtained at room temperature previous to the humidity test by more than the tolerance specified in the detail specification; the friction errors shall not exceed the tolerance for room temperature by more than that specified in the detail specification; and there shall be no insulation breakdown as a result of the dielectric strength test.

4.6.18 Thermal shock. This test is applicable only to Hermetically sealed indicators. The indicator shall be subjected to eight cycles of immersion in water at a temperature of 85° $\pm 2^{\circ}$C and then in water at a temperature of 5° $\pm 2^{\circ}$C without evidence of moisture penetration or damage to coating or the case. The length of time for each immersion shall be 30 minutes, with not more than 5 seconds between immersions. After eight continuous cycles have been completed, the indicator shall be subjected to the test specified in 4.6.6. Leakage in excess of that permitted in 4.6.6 or fogging of the cover glass inside the case shall be cause for rejection.

4.6.19 Fungus. The indicator shall be subjected to a fungus test IAW

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MIL-STD-810, method 508, procedure I. The unit selected shall not have been previously subjected to the salt fog test. At the end of the test period, the indicator shall be carefully examined to determine that no fungus growth has occurred. This test is applicable only to nonhermetically sealed indicators and shall be conducted at the conclusion of all other tests except the salt fog test.

4.6.20 Salt fog. The indicator shall be subjected to a salt fog test IAW MIL-STD-810, method 509, procedure I. There shall be no evidence of external corrosion or deterioration that might affect subsequent operation. The indicator shall then be subjected to and shall pass the test specified in 4.6.2. The external surface of the indicator may be washed with distilled water and air dried prior to the dielectric strength portion of this test.

4.6.20.1 Salt fog test. At the completion of the salt fog test, the indicator shall be subjected to the test specified 4.6.4. The scale errors shall not differ from the readings at room temperature before the salt fog test by more than the tolerance specified in the detail specification. Hermetically sealed indicators shall be subjected to and shall pass the test specified in 4.6.6. Corrosion of the electrical receptacle shell or pins shall be cause for rejection.

4.6.21 Electromagnetic interference. The indicator shall be subjected to an electromagnetic interference test IAW MIL-STD-462 within the limits of 0.15 and 150 MHz.

4.6.22 Unit torque gradient. The indicator shall be connected to the master transmitter synchro and the master transmitter locked in the electrical zero position. The synchro rotors shall then be energized with rated voltage and frequency ±1 percent. Weights shall be applied to a torque pulley which is positioned on the rotor shaft to rotate the shaft in a ccw direction through an angle of 5 degrees of arc. The torque tending to return the shaft to its original position shall be computed and shall be not less than 2,000 milligram-millimeters per degree of displacement.

4.6.23 Life, contrast, and lamp circuit. These tests shall be conducted IAW MIL-L-25467 or MIL-L-27160, as applicable.

4.7 Inspection of preparation for delivery. Preservation, packaging, packing, and marking shall be examined to determine compliance with section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level "A" or "C", as specified (see 6.2)

5.1.1 Preservation-packaging. Unless otherwise specified by the contracting activity, item shall be packaged in quantity unit pack (QUP) of one each. Each nonhermetically sealed item will be provided a preservation method IIc IAW MIL-P-116. Each hermetically sealed item will be provided a preservation method IA-8 IAW MIL-P-116.

5.1.2 Cleaning. Item shall be cleaned IAW MIL-P-116 C-1 process.

5.1.3 Drying. Immediately after cleaning, the item shall be dried following

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any one or combination of the drying procedures listed in MIL-P-116. The drying procedures employed shall not be injurious to the item.

5.1.4 Level A. Item shall be preserved IAW MIL-P-116 and MIL-STD-2073-1A to provide a method IIc IAW MIL-P-116 for nonhermetically sealed items and method IA-8 for hermetically sealed items.

5.1.5 Level C. Items shall be individually preserved in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity as specified in MIL-P-116 and MIL-STD-2073-1B.

5.2 Packing. Packing shall be level "A", "B", or "C", as specified (see 6.2).

5.2.1 Container-cushioning. Unit container requirements shall conform to MIL-B-131, barrier material, unless otherwise specified by the contracting activity. Intermediate container requirements shall conform to PPP-B-636 or PPP-B-1672, weather-resistant, unless otherwise specified by the contracting activity. Intermediate containers shall be large enough to allow for application of sufficient cushioning material, between the unit container and intermediate container. Cushioning material shall be of sufficient density and thickness to insure shock transmission does not exceed peak values in G's established for the item.

5.2.2 Level A. Item will be preserved as specified in 5.1 and shall be packed in exterior containers conforming to PPP-B-636 or PPP-B-1672, weather-resistant, unless otherwise specified by the contracting activity. Exterior container shall be uniform shape, size and minimum tare and cube, consistent with the protection required. Closure shall be IAW appropriate PPP-B-636 or PPP-B-1672 procedure, as specified by contractor, special packaging instructions (SPI), and contracting activity.

5.2.3 Level B. Same as Level "A" unless otherwise specified by SPI or the contracting activity.

5.2.4 Level C. Item will be packed in containers in such a manner as to afford adequate protection against physical damage during direct domestic shipment from the supply source to the first receiving activity. These packs shall conform to MIL-STD-2073-1B.

5.3 Marking. Unit, intermediate, and exterior containers shall be marked IAW special markings required by the contractor, SPI, and MIL-STD-129.

5.4 Inspection and test. Test of methods of preservation shall be accomplished IAW section 4 of MIL-P-116 to insure compliance with section 5 of this specification. Packaging tests shall be conducted IAW rough handling as specified in FED-STD-101.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The self-synchronous indicators covered by this specification are intended for use in providing remote indication of aircraft functions.

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6.2 *Acquisition requirements*. *Acquisition documents must specify the following:*

a. Title, number, and date of this specification
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1)
c. Whether receptacles will be in accordance with table I or II (see 3.5.4)
d. Whether internal wiring will be in accordance with figure 3, 4, or 5 (see 3.8)
e. When sampling plan B tests will not be conducted (see 4.5.2.2)
f. Levels of preservation, packaging, and packing required (see section 5).

6.3 *Definitions*

6.3.1 Amplitude. Amplitude is the extent of motion as measured from one extreme to the opposite extreme.

6.3.2 Hermetic seal. A hermetic seal is a perfectly closed and airtight seal made between vitric or metallic materials, or both. A hermetic seal does not include seals accomplished by gaskets.

6.4 Subject term (key word) listing.

Single phase
Synchro indicators
26 or 115 volt

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
AIR FORCE -99
AIR FORCE -11
ARMY -AV
NAVY -AS

User Activities:
NAVY - MC, CGM, SH

Preparing Activity:
AIR FORCE -71

Project number:
6620-0501

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INSTRUCTIONS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-I-7037D	2. DOCUMENT DATE (YYMMDD) 15 December 1992

3. DOCUMENT TITLE
Indicators, Synchro, Aircraft, General Specification

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198-290

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